

Counterfactual Graph for Multi-Agent LLM Calibration

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Abstract

Multi-agent LLM systems often treat agreement as evidence: when many agents in a panel give the same answer, that answer is assumed to be more reliable. We show that this assumption can fail after agents communicate. Communication can induce correlated failures and false consensus, so the same vote share may reflect reliable agreement in one topology but overconfidence in another. We propose CAGE-CAL, a counterfactual agent-graph calibration framework for multi-agent LLMs. For each query, CAGE-CAL compares an observed post-communication agent graph with a matched counterfactual no-communication graph, capturing both pairwise failure correlations and group-level dependencies. Rather than simply counting how many agents agree, CAGE-CAL estimates the counterfactual shift between observed and no-communication dependence, and calibrates confidence accordingly. Across five benchmarks, CAGE-CAL improves reliability discrimination with competitive ECE, and its calibrated confidence further improves topology selection over the best fixed-topology strategy. Our code repo is available [here](#).

1 Introduction

Multi-agent LLM systems are increasingly used in agentic pipelines such as code review, search augmentation, and clinical triage (Du et al., 2024; Hong et al., 2024; Wu et al., 2024). In these settings, systems not only answer questions, but also indicate how much their answers should be trusted. This confidence signal directly affects downstream decisions: high-confidence answers may be accepted automatically, whereas low-confidence answers may be deferred to human review. Confidence should therefore be calibrated to the true probability of correctness, making calibration not merely an evaluation metric but a prerequisite for reliable and safe deployment (Guo et al., 2017; Geifman and El-Yaniv, 2017).

Unlike single-agent calibration, which characterizes the uncertainty of one model on one query (Zhou et al., 2025; Bani-Harouni et al., 2026), multi-agent calibration must account for a joint decision process in which multiple agents’ outputs are shaped by communication topology (Huang et al., 2026; Zhang et al., 2025c; Shi et al., 2026). This setting introduces several distinct challenges. 1) *Dependence-blind calibration*. Existing multi-agent calibration methods typically rely on uncertainty scores derived from the aggregate system output, such as vote share, entropy, or semantic disagreement (Wang et al., 2023b; Kadavath et al., 2022; Feng et al., 2025; Jiang, 2026; Chen et al., 2026). By treating agent responses as independent samples and compressing disagreement into a single scalar, these methods discard the structural information needed to distinguish benign disagreement from harmful agreement. 2) *Heterogeneous-agent calibration*. Existing multi-agent calibration methods primarily study agents instantiated from a shared LLM policy (Zhu et al., 2026; Qiao et al., 2026). In practice, however, panels¹ may include different LLMs with distinct knowledge boundaries, confidence scales, and failure modes. Calibrating confidence for such heterogeneous panels remains underexplored. 3) *Topology generalization*. Existing calibration methods are largely designed and validated in all-to-all communication multi-agent systems (Zhu et al., 2026; Qiao et al., 2026). Practical multi-agent systems, however, may use diverse communication topologies, including independent voting, chains, hub-spoke structures, and trees. Whether calibration methods developed for debate-style agents generalize to these more diverse and structurally complex settings remains unclear.

To address these challenges, we first analyze how agreement and disagreement can become mis-

¹A panel refers to the group of agents whose responses are aggregated to produce a final answer.

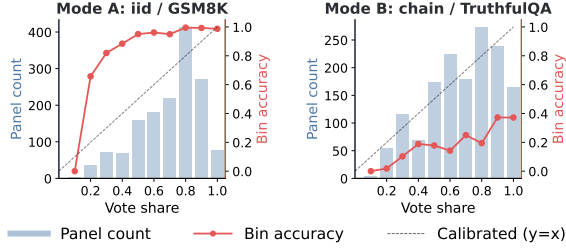


Figure 1: **The two failure modes.** Vote-share histogram (blue) vs. per-bin panel accuracy (red). The dashed line denotes perfect calibration. In iid/GSM8K, vote share underestimates accuracy (under confidence), while in chain/TruthfulQA, it overestimates accuracy (over confidence).

leading confidence signals in both homogeneous and heterogeneous panels across different communication topologies. This analysis reveals two opposing failure modes. (1) *Diversity-induced under-confidence* (DUC): incorrect agents often spread across different wrong answers, so vote share underestimates panel reliability. (2) *Communication-induced over-confidence* (COC): agents may converge to a wrong answer through influence, so vote share overestimates reliability.

Based on this diagnosis, we then propose CAGE-CAL (Counterfactual Agent-Graph Calibration), a confidence calibration framework for heterogeneous multi-agent LLMs across diverse communication topologies. For each query, CAGE-CAL compares the observed communication graph with an IID counterfactual graph, where the same agents vote independently without communication. Their difference captures communication-induced dependence, enabling the calibrator to down-calibrate COC cases where high agreement reflects agent coupling rather than independent evidence. To address DUC, CAGE-CAL further combines pairwise failure correlations with high-order hyperedges over model family, prompt role, answer cluster, and topology. These group-level signals reveal when disagreement is benign, e.g., when wrong agents are dispersed across weak answer groups. Together, these components adjust confidence in opposite directions: lowering confidence for false consensus and preserving or increasing confidence for harmless diversity.

Our contributions are summarized as follows:

- We analyze multi-agent calibration across different LLMs and communication topologies, identifying two failure modes: DUC and COC.
- We propose CAGE-CAL, a new multi-agent

confidence calibration framework that calibrates confidence by contrasting communication graphs with IID counterfactual graphs and modeling failure dependencies.

- We evaluate CAGE-CAL on five benchmarks and topologies, showing improved reliability discrimination with competitive calibration error. We further introduce CAGE-SELECT, which uses calibrated confidence to select the most reliable topology and improve final panel accuracy.

2 Related Work

Multi-Agent Uncertainty Quantification. Existing uncertainty estimators for LLM panels often summarize the answer distribution with a scalar score (Wang et al., 2023b; Kadavath et al., 2022; Feng et al., 2025; Chen et al., 2026; Jiang, 2026). For example, DiscoUQ-LLM (Jiang, 2026) learns from disagreement features, while MATU (Chen et al., 2026) models uncertainty through tensor decomposition over multi-agent rollouts (Jiang, 2026; Chen et al., 2026). These methods are useful when disagreement reflects uncertainty. However, disagreement can be misleading: benign diversity may cause under-confidence, while communication-induced agreement may cause over-confidence. CAGE-CAL therefore conditions confidence on agent dependencies rather than disagreement alone.

Multi-Agent Confidence Calibration. Existing work can be broadly grouped into three lines. The first line, *Post-hoc plurality calibrators* (Zhang et al., 2026), calibrates scalar uncertainty signals, such as vote share or entropy, using post-hoc methods (Platt et al., 1999; Zadrozny and Elkan, 2002; Kumar et al., 2019). The second line studies *LLM-elicited confidence estimators* (Lin et al., 2022a; Yang et al., 2024), where LLMs are trained or prompted to report calibrated confidence. For example, LLM-Cal (Lin et al., 2022a) fine-tunes an LLM to align verbalized confidence with empirical correctness. The third line develops *trained calibrators* (Li et al., 2025; Jiang, 2026; Zhu et al., 2026), which estimate final-answer correctness from confidence and disagreement features. For example, DiscoUQ-LLM (Jiang, 2026) uses logistic regression over vote confidence and hand-crafted disagreement features. Despite their effectiveness, these methods largely overlook communication-induced dependencies, heterogeneous LLM panels,

and diverse communication topologies.

3 Problem Statement

A multi-agent panel consists of N language model agents, a communication topology represented by adjacency matrix W_c , and a communication protocol. For a query x , each agent emits a final answer $y_i(x)$, and the panel produces one prediction $\hat{y}(x) = \text{plurality}(\{y_i(x)\}_{i=1}^N)$, where plurality vote selects the most frequent normalized answer. Given the panel prediction $\hat{y}(x)$, our task is to estimate the calibrated probability that this panel answer is correct:

$$\hat{p}(x) = \Pr(C(x) = 1 \mid \mathcal{M}(x), W_c), \quad (1)$$

where $C(x) = \mathbf{1}[\hat{y}(x) = y^*(x)]$ denotes whether the panel prediction is correct. Here, $\mathcal{M}(x)$ denotes the observable final panel state, including agents' final answers, answer-level confidence scores, vote statistics, and answer embeddings. For later use, we also define the agent-level correctness indicator $c_i(x) = \mathbf{1}[y_i(x) = y^*(x)]$, which denotes whether agent i answers correctly.

4 Calibration Failure Modes

To diagnose calibration failures in multi-agent systems, we analyze panel behavior across a grid of communication topologies and benchmarks and identify two recurring failure modes: Diversity-Induced Under-Confidence (DUC) and Communication-Induced Over-Confidence (COC). Figure 1 illustrates these modes with two representative cells: iid on GSM8K and chain on TruthfulQA. We defer the full analysis pipeline to Appendix B.

Failure mode A: DUC. Let $v_a(x)$ denote the fraction of agents that output answer a for query x : $v_a(x) = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[y_i(x) = a]$. Let $v(x) = v_{\hat{y}(x)}(x)$ be the vote share of the plurality answer $\hat{y}(x)$. For a vote-share bin centered at s , define $\mathcal{B}_s = \{x : v(x) \in [s - \epsilon, s + \epsilon]\}$. DUC occurs when examples in this bin are empirically more accurate than their vote share suggests, while competing answers remain weakly supported:

$$\begin{aligned} \frac{1}{|\mathcal{B}_s|} \sum_{x \in \mathcal{B}_s} C(x) &> s, \\ \frac{1}{|\mathcal{B}_s|} \sum_{x \in \mathcal{B}_s} \max_{a \neq \hat{y}(x)} v_a(x) &\ll s. \end{aligned} \quad (2)$$

Failure mode B: COC. COC occurs when examples in a high-vote-share bin are empirically less accurate than their vote share suggests, while the agents supporting the plurality answer are strongly dependent:

$$\begin{aligned} \frac{1}{|\mathcal{B}_s|} \sum_{x \in \mathcal{B}_s} C(x) &< s, \\ \frac{1}{|\mathcal{B}_s|} \sum_{x \in \mathcal{B}_s} \text{Dep}(S_{\hat{y}}(x)) &\gg 0, \end{aligned} \quad (3)$$

where $S_{\hat{y}}(x) = \{i : y_i(x) = \hat{y}(x)\}$ is the set of agents supporting the plurality answer, and $\text{Dep}(S_{\hat{y}}(x))$ measures the average dependence among these supporting agents.

5 CAGE-CAL: A Counterfactual Graph Calibration Framework

5.1 Motivation and Approach Overview

Based on the analysis in Section 4, we find that the same vote pattern can imply opposite reliability. Benign diversity disperses wrong votes and makes reliable panels appear uncertain, while communication can concentrate agents on the same wrong answer and create false confidence. This motivates CAGE-CAL, which calibrates confidence by modeling the dependency structure behind agent agreement. Figure 2 (A) illustrates the two motivating failure modes.

Figure 2 (B-C) provides an overview of the CAGE-CAL framework, which consists of three components. First, *Agent Dependency Graph Construction* builds two instance-level graphs: a counterfactual IID graph G_x^0 , which captures dependencies that would exist without communication, and a post-communication graph G_x^T , which captures dependencies after agents interact under topology T . Second, *Counterfactual Shift Calibration*, which includes 1) *Calibration Agent Graph Encoder* and 2) *Counterfactual Shift Calibration Learning*, uses the difference between these two graphs as a calibration signal to determine whether agreement reflects independent evidence or correlated failure. Finally, we introduce CAGE-SELECT (D), which uses the calibrated confidence from CAGE-CAL to select the most reliable topology for each query and improve final panel accuracy.

5.2 Agent Dependency Graph Construction

For each query x and communication topology T , CAGE-CAL constructs two graphs over matched

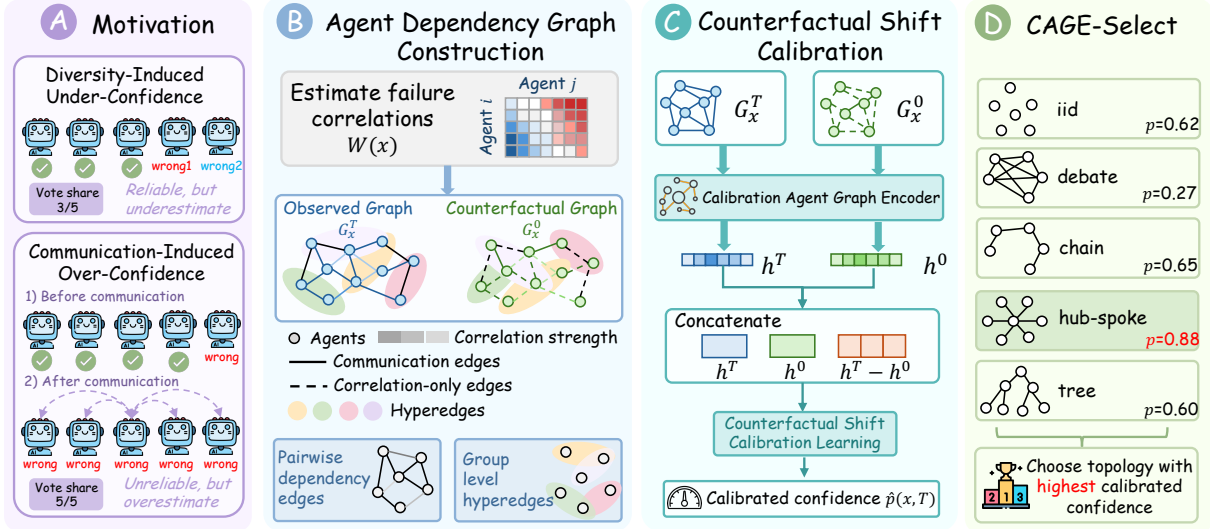


Figure 2: **Overview of CAGE-CAL.** The same vote share can reflect different reliability depending on how agreement is formed. CAGE-CAL constructs matched observed and IID counterfactual dependency graphs, encodes pairwise and group-level dependencies, and uses the counterfactual shift to estimate calibrated panel confidence. The resulting confidence can also guide topology selection.

agent identities. The observed graph G_x^T is built from the completed panel under topology T . The counterfactual graph G_x^0 is built from the cached iid run of the same query and rollout, where the same agents answer without communication. We keep the shared agent identities between the two runs, so that the two graphs describe the same panel under two relational states. This matching makes the contrast between G_x^T and G_x^0 reflect communication induced changes rather than changes in the agent population.

Note. Each node represents one agent. For both G_x^T and G_x^0 , we use the same 23-dimensional node feature schema:

$$v_i(x) = [s_i, p_i, r_i, \mu_i, \sigma_i^2, m_i, n^{-1}, a_i], \quad (4)$$

where s_i is the agent confidence score. p_i indicates whether the agent supports the panel plurality answer, r_i is the normalized vote rank of the agent’s answer, (μ_i, σ_i^2, m_i) are local correlation summaries from the corresponding row of the instance conditional correlation matrix, n^{-1} is the inverse panel size, and a_i is a low dimensional answer embedding. Further details on node feature construction are provided in Appendix C.1.

Pairwise dependency edges. Pairwise edges describe local dependence between agents. For agents i and j , the edge feature in the observed graph is $e_{ij}^T(x) = [W_{c,T}^{ij}, W_T^{ij}(x)]$, where $W_{c,T}^{ij}$ is the binary communication adjacency under topology T and records who could directly observe or influence

whom under the topology. $W_T^{ij}(x)$ is the instance-conditional correctness correlation estimated from nearby training queries, indicating whether agents i and j tend to succeed or fail together on queries similar to x . Further details about this correlation estimation are provided in Appendix C.2. In the IID counterfactual graph, the communication-adjacency feature is set to zero, since no communication occurs among agents.

Group level hyperedges. Pairwise edges capture only dyadic dependence, while many sources of dependence act at the group level. We therefore augment each graph with a hyperedge set

$$\mathcal{H}_x^T = \mathcal{H}_{\text{fam}} \cup \mathcal{H}_{\text{role}} \cup \mathcal{H}_{\text{ans}}(x) \cup \mathcal{H}_{\text{topo}}^T. \quad (5)$$

Here, $\mathcal{H}_{\text{fam}}$ connects agents sharing the same backbone family, $\mathcal{H}_{\text{role}}$ connects agents using the same prompting role, $\mathcal{H}_{\text{ans}}(x)$ connects agents that emit the same normalized answer for query x , and $\mathcal{H}_{\text{topo}}^T$ connects agents exposed through topology T . The IID counterfactual graph does not include this communication exposure hyperedge. We create a hyperedge only when it contains at least two agents. These hyperedges are dependence units rather than auxiliary metadata. A shared backbone can induce shared blind spots. A shared prompting role can induce similar reasoning patterns. An answer cluster can become a consensus unit. A communication exposure group can reflect shared influence from the topology. This design follows the empirical analysis in Appendix B.1: multi-agent reliability

is shaped by both population side structure and topology induced coupling (Li et al., 2026b).

5.3 Calibration Agent Graph Encoder

Given the matched graphs G_x^T and G_x^0 , CAGE-CAL uses the calibration agent graph encoder f_θ to obtain:

$$h^T = \text{Pool}(f_\theta(G_x^T)), \quad h^0 = \text{Pool}(f_\theta(G_x^0)). \quad (6)$$

For each graph, the encoder updates every agent representation with two relational signals. The pairwise update aggregates over agent-agent edges:

$$p_i^{(\ell)} = \sigma \left(\sum_{j \in \mathcal{N}(i)} \alpha_{ij}^{(\ell)} W_p^{(\ell)} z_j^{(\ell)} \right), \quad (7)$$

where $\alpha_{ij}^{(\ell)}$ is the attention weight from agent j to agent i at layer ℓ . It determines how much information agent i receives from agent j . The weight is computed using the two agent representations and their edge feature $e_{ij}(x) = [W_c^{ij}, W_{ij}(x)]$, allowing communication exposure and query-specific failure correlation to condition message passing. Here, $z_i^{(\ell)}$ denotes the layer- ℓ representation of agent i , $W_p^{(\ell)}$ is a learnable projection matrix. The group-level update aggregates over hyperedges:

$$q_i^{(\ell)} = \sigma \left(\sum_{e \in \mathcal{H}(i)} \frac{1}{d_i} \frac{1}{|e|} \sum_{j \in \mathcal{V}(e)} W_q^{(\ell)} z_j^{(\ell)} \right), \quad (8)$$

where $\mathcal{H}(i)$ is the set of hyperedges containing agent i , $d_i = |\mathcal{H}(i)|$ is the hyperedge degree of agent i , and $|e|$ is the number of agents in hyperedge e . This is equivalent to an unweighted degree-normalized hypergraph convolution using the node-hyperedge incidence matrix H . The operation first aggregates node representations into group representations and then broadcasts the group information back to member agents. Here, $W_q^{(\ell)}$ is a learnable projection matrix. The two updates are then fused by concatenation:

$$z_i^{(\ell+1)} = \phi \left(p_i^{(\ell)} \| q_i^{(\ell)} \right), \quad (9)$$

where ϕ denotes dropout applied after concatenation. Because the same encoder is used for both graphs, h^T and h^0 lie in the same representation space. We define the communication-induced shift and the final calibration representation as

$$\Delta h_x^T = h^T - h^0, \quad z_x = [h^T \| h^0 \| \Delta h_x^T \| b_x]. \quad (10)$$

Here b_x is a benchmark one-hot vector. The three graph terms serve different roles. The observed representation h^T describes the panel that produced the final answer. The counterfactual representation h^0 describes the background dependence of the same agents when they answer independently. The difference $h^T - h^0$ captures the topology-induced shift from independent answering to communication.

5.4 Counterfactual Shift Calibration Learning

We train with a calibration-aware objective combining smoothed binary cross-entropy and a Brier penalty:

$$\mathcal{L} = \text{BCE}(\hat{p}(x), \tilde{C}(x)) + \lambda_{\text{Brier}} (\hat{p}(x) - C(x))^2 \quad (11)$$

where $\hat{p}(x) = \sigma(\text{MLP}(z_x))$ is the predicted probability that the panel answer is correct. The binary target is $C(x) = \mathbf{1}[\hat{y}(x) = y^*(x)]$, where $\hat{y}(x)$ denotes the panel prediction and $y^*(x)$ denotes the gold answer. To regularize the binary objective, we use the smoothed target $\tilde{C}(x) = C(x)(1 - 2\alpha) + \alpha$. The BCE term encourages discrimination between correct and incorrect panel predictions, while the Brier term directly penalizes miscalibrated probability estimates.

6 Experiments

6.1 Experimental Setup

We use the 25-cell grid from Section 3, formed by five benchmarks and five communication topologies. The benchmarks are TriviaQA (Joshi et al., 2017), TruthfulQA (Lin et al., 2022b), MMLU-Pro (Wang et al., 2024), GSM8K (Cobbe et al., 2021), and BIG-Bench Hard (Suzgun et al., 2023). The topologies are iid, debate, chain, hub-spoke, and tree. Each topology and benchmark cell is evaluated with three rollouts. For each query, the panel first produces a plurality answer, and the calibrator then scores the reliability of that answer. Metrics are averaged over topologies and rollouts within each benchmark. The Mean column macro-averages the five benchmarks. For CAGE-SELECT, we evaluate only matched test groups where all five topology outputs are available for the same query. Full experimental details are provided in Appendix D.

Method	TriviaQA		TruthfulQA		MMLU-Pro		GSM8K		BBH		Mean	
	ECE↓	AUROC↑	ECE↓	AUROC↑	ECE↓	AUROC↑	ECE↓	AUROC↑	ECE↓	AUROC↑	ECE↓	AUROC↑
<i>Post-hoc plurality calibrators</i>												
Plurality share	11.43 ± 0.07	81.89 ± 0.79	25.23 ± 0.89	71.16 ± 3.89	11.70 ± 0.89	61.78 ± 0.80	17.98 ± 0.22	83.01 ± 2.22	23.87 ± 1.36	66.15 ± 1.49	18.04 ± 0.21	72.80 ± 1.52
+ Platt	7.96 ± 1.64	81.89 ± 0.79	9.82 ± 0.15	71.16 ± 3.89	9.96 ± 0.67	61.78 ± 0.80	2.86 ± 0.23	83.01 ± 2.22	10.20 ± 0.90	66.15 ± 1.49	8.16 ± 0.36	72.80 ± 1.52
+ Isotonic	8.11 ± 1.45	81.90 ± 0.73	8.47 ± 0.00	70.45 ± 3.99	10.83 ± 1.51	61.00 ± 0.47	3.22 ± 0.60	80.00 ± 2.87	7.96 ± 0.09	65.37 ± 2.15	7.72 ± 0.73	71.74 ± 1.75
+ Scaling-bin.	9.58 ± 0.19	81.30 ± 0.23	8.41 ± 1.88	70.83 ± 3.62	7.79 ± 1.04	61.97 ± 0.86	2.28 ± 0.34	82.85 ± 1.43	<u>6.86 ± 2.42</u>	65.95 ± 1.29	<u>6.98 ± 0.07</u>	72.58 ± 1.39
<i>LLM-elicited confidence estimators</i>												
LLM-Cal (no topo)	7.99 ± 0.36	85.74 ± 1.17	54.88 ± 0.83	78.50 ± 1.54	33.19 ± 2.36	57.64 ± 1.22	8.96 ± 0.32	67.18 ± 1.09	13.87 ± 0.85	63.92 ± 0.12	23.78 ± 0.33	70.60 ± 0.49
LLM-Cal (+topo)	8.71 ± 0.07	85.73 ± 0.01	54.52 ± 0.87	78.48 ± 2.25	32.03 ± 1.00	58.33 ± 0.62	8.28 ± 0.57	71.16 ± 5.16	14.40 ± 1.89	62.56 ± 0.80	23.59 ± 0.11	71.25 ± 1.52
Collab. Cal.	8.84 ± 0.73	86.30 ± 0.45	52.87 ± 0.95	78.24 ± 1.16	31.30 ± 1.84	59.25 ± 2.35	8.01 ± 1.50	70.01 ± 2.72	11.59 ± 0.08	64.71 ± 0.48	22.52 ± 0.13	71.70 ± 0.30
<i>Trained calibrators</i>												
Scalar + GBT	9.13 ± 1.05	79.24 ± 0.78	10.28 ± 0.81	66.81 ± 2.30	18.38 ± 0.76	63.71 ± 2.04	4.34 ± 0.27	76.35 ± 1.86	10.89 ± 1.63	69.68 ± 0.33	10.60 ± 0.18	71.16 ± 0.28
GraphCal	11.05 ± 0.48	80.06 ± 1.00	33.66 ± 1.18	63.75 ± 3.20	13.33 ± 0.81	55.18 ± 0.47	19.70 ± 0.32	81.02 ± 1.12	19.77 ± 0.55	<u>71.53 ± 0.15</u>	19.50 ± 0.20	70.31 ± 0.28
DiscoUQ-LLM	6.88 ± 1.59	82.19 ± 0.11	<u>6.99 ± 1.17</u>	71.93 ± 4.48	10.55 ± 0.66	<u>64.05 ± 0.76</u>	2.43 ± 0.70	83.19 ± 3.28	8.54 ± 1.03	65.93 ± 1.27	7.08 ± 0.49	<u>73.46 ± 1.94</u>
CAGE-CAL (ours)	4.25 ± 1.49	<u>86.12 ± 0.78</u>	6.50 ± 0.72	79.66 ± 2.43	<u>9.30 ± 1.62</u>	77.74 ± 0.75	1.64 ± 0.05	84.04 ± 3.55	6.12 ± 2.57	90.48 ± 0.81	5.56 ± 0.03	83.61 ± 1.34[†]

Table 1: **In-distribution headline calibration results.** ECE (↓) and AUROC (↑), mean ± std over 3 rollout seeds, averaged across the five topologies. Mean column macro-averages benchmarks. **Bold** marks the best result in each column, and underline marks the second best. †: Wilcoxon $p < 0.05$ vs DiscoUQ-LLM on Mean AUROC. Heuristic UQ baselines and AUARC / Brier for all methods are deferred to Table 7 and Table 8.

6.2 Baselines and Metrics

We compare CAGE-CAL against three categories of baselines to address different evaluation dimensions: *Post hoc plurality calibrators*: Calibrate the confidence score computed from plurality share (Kuncheva, 2004) on the validation split using three post-hoc methods: (1) Platt scaling (Platt et al., 1999), (2) Isotonic regression (Zadrozny and Elkan, 2002), and (3) Scaling-binning (Kumar et al., 2019); *LLM-elicited confidence estimators*. These baselines use an LLM judge to elicit a probability-like confidence score for the panel prediction. They test whether panel reliability can be inferred from the final agent responses, optionally augmented with the topology description. The baselines include: (1) LLM-Cal without topology information, (2) LLM-Cal with topology information, and (3) Collaborative Calibration (Yang et al., 2024); *Trained calibrators*. (1) Scalar + GBT (Ke et al., 2017) uses vote, confidence, and graph-summary features without relational encoding; (2) GraphCal (Li et al., 2025) adapts graph-based calibration to the panel setting; and (3) DiscoUQ-LLM (Jiang, 2026) serves as a strong baseline based on disagreement features. We also evaluate answer entropy (Kuhn et al., 2023), average log probability (Kadavath et al., 2022), DiverseAgentEntropy (Feng et al., 2025), and MATU (Chen et al., 2026). These methods provide uncertainty scores rather than calibrated probabilities, so we report AUROC and AUARC without interpreting the scores probabilistically.

Metrics. Our primary metrics are ECE and AUROC. ECE measures whether predicted confidence matches empirical correctness, while AUROC mea-

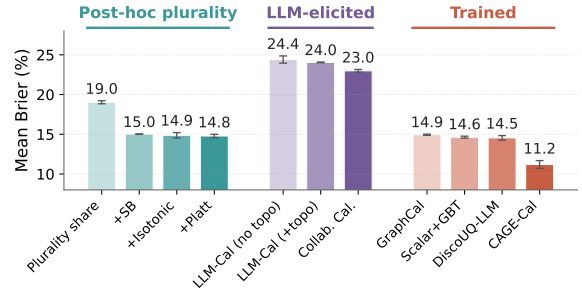


Figure 3: **Mean Brier score by method family** (lower is better). Within each family, bars are sorted worst → best (light → dark). CAGE-CAL (rightmost) has the lowest Brier overall.

asures whether correct panel answers receive higher scores than incorrect ones. We additionally report Brier score and AUARC as complementary metrics. Brier score evaluates probability quality under a proper scoring rule, and AUARC evaluates selective prediction when panel answers are accepted in decreasing confidence order. For ranking-only UQ scores, we omit ECE unless a probability scale is introduced through the shared validation-set calibration protocol.

6.3 Main Calibration Results

Table 1 reports the headline in-distribution results. The first block shows that post-hoc calibration can substantially reduce the ECE of plurality share, from 18.04 to 6.98 with scaling-binning. However, these methods still rely on the same scalar agreement signal. Platt scaling is monotonic and therefore preserves the AUROC of plurality share at 72.80; isotonic regression and scaling-binning only change ranking slightly through ties and bins. This confirms that post-hoc calibration can improve probability scale, but cannot add the struc-

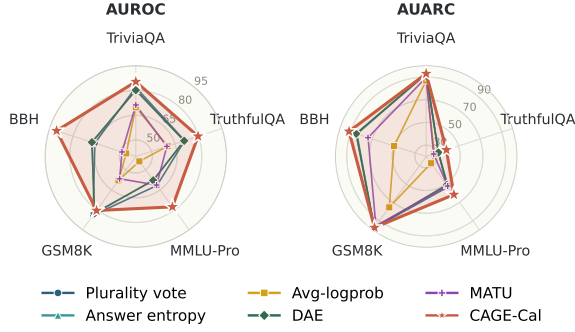


Figure 4: **AUROC and AUARC of CAGE-CAL vs. heuristic UQ baselines.** Each axis reports benchmark-level mean over 3 rollouts and 5 topologies. Per-benchmark numbers in Appendix Table 7.

tural information needed to distinguish independent agreement from communication-induced consensus. LLM-elicited confidence estimators are also limited in this setting. Their mean ECE remains high, ranging from 22.52 to 23.78, and adding topology information to the prompt brings only a small AUROC change. This suggests that a topology label alone is not enough for a judge model to recover the query-specific dependence among agents. Among trained calibrators, CAGE-CAL achieves the best mean ECE and the strongest mean AUROC. Compared with DiscoUQ-LLM, the strongest prior trained baseline, CAGE-CAL reduces mean ECE from 7.08 to 5.56 and improves mean AUROC from 73.46 to 83.61. The AUROC gain is especially large on MMLU-Pro and BBH, where correlated agreement is more harmful: CAGE-CAL improves AUROC by 13.69 points on MMLU-Pro and 24.55 points on BBH over DiscoUQ-LLM. These gains support the central claim that panel confidence should depend not only on how many agents agree, but also on how that agreement was formed. Figure 3 provides a complementary view through Brier score. CAGE-CAL achieves the lowest mean Brier score at 11.2, outperforming DiscoUQ-LLM (14.5), Scalar + GBT (15.6), and GraphCal (19.2). Thus, the improvement is not only an artifact of ECE binning; CAGE-CAL also produces sharper and better scaled probability estimates.

6.4 Ranking-Only UQ Comparison

Figure 4 compares CAGE-CAL with ranking-only UQ baselines using AUROC and AUARC. These methods provide useful uncertainty scores, but they do not define a calibrated probability scale. Plurality vote, entropy, average log probability, DiverseAgentEntropy, and MATU all summarize the

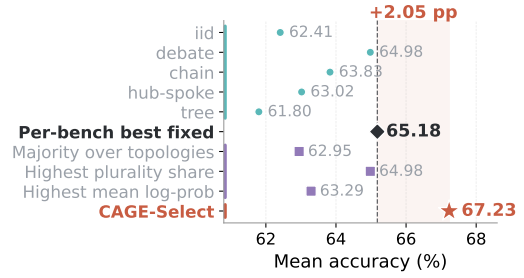


Figure 5: **Mean accuracy of routing strategies.** Dashed line marks per-bench best fixed (65.18). Per-bench breakdown in Appendix Table 9.

panel with scalar signals. Such signals work when disagreement reflects uncertainty, but they miss the difference between benign diversity and false consensus. CAGE-CAL gives the strongest overall reliability ranking across the five benchmarks. The advantage is most visible on BBH and MMLU-Pro, where scalar disagreement signals are less reliable. This aligns with the failure-mode analysis: when agents become correlated through shared model families or communication paths, the reliability of the final answer depends on the dependency structure behind the vote, not only on the vote distribution itself.

6.5 Confidence-Routed Topology Selection

We further test whether calibrated confidence can serve as a control signal for multi-agent inference. Instead of committing to a single topology for all queries, CAGE-SELECT uses CAGE-CAL confidence to choose which topology output should be trusted for each query. Figure 5 shows that the best fixed topology reaches 65.18% mean accuracy, while simple routing rules based on plurality share or mean log probability do not improve over it. CAGE-SELECT reaches 67.23%, a +2.05 point gain. Thus, CAGE-CAL confidence is not only calibrated within a topology, but also comparable across topologies, making it useful as a control signal for multi-agent inference.

7 Analysis

7.1 Component Ablation

Table 3 ablates the main components of CAGE-CAL. Scalar-summary baselines compress each panel into fixed statistics and remain far below graph-based variants, with the stronger GBT head reaching only 74.78 AUROC and 72.70 AUARC. The observed graph encoder raises AUROC to 81.25, showing that panel reliability depends on relational structure beyond aggregate vote and

Method	TriviaQA		TruthfulQA		MMLU-Pro		GSM8K		BBH		Mean	
	ECE↓	AUROC↑	ECE↓	AUROC↑	ECE↓	AUROC↑	ECE↓	AUROC↑	ECE↓	AUROC↑	ECE↓	AUROC↑
Scalar + GBT	11.15 ± 4.43	80.28 ± 4.95	23.50 ± 12.68	70.46 ± 2.14	16.39 ± 4.04	60.73 ± 4.28	10.14 ± 3.84	80.58 ± 2.49	13.33 ± 4.22	62.48 ± 2.54	14.90 ± 1.73	70.91 ± 2.54
GraphCal	11.93 ± 4.66	80.13 ± 5.01	34.15 ± 9.54	63.51 ± 2.89	14.41 ± 2.17	54.79 ± 5.86	19.59 ± 6.43	80.92 ± 6.54	20.05 ± 4.09	70.62 ± 3.60	20.03 ± 1.87	70.00 ± 2.40
DiscoUQ-LLM	12.52 ± 5.24	82.60 ± 4.90	24.42 ± 13.93	71.78 ± 3.72	16.83 ± 4.67	59.37 ± 7.95	10.09 ± 4.12	84.59 ± 3.95	15.25 ± 4.40	63.00 ± 3.58	15.82 ± 8.45	72.27 ± 11.32
CAGE-CAL (ours)	4.63 ± 0.71	85.74 ± 1.41	8.70 ± 1.84	79.59 ± 4.67	11.19 ± 1.29	76.67 ± 3.63	1.89 ± 1.31	80.39 ± 5.75	6.38 ± 1.38	88.67 ± 1.12	6.56 ± 0.68	82.21 ± 1.84

Table 2: **Leave-one-topology-out (LOTO) generalization.** ECE (↓) and AUROC (↑), mean over the 5 held-out-topology folds. Trained calibrators only.

Variant	ECE ↓	AUROC ↑	AUARC ↑
<i>Scalar-summary baselines</i>			
Scalar summaries + LR	8.03±0.35	74.12±1.24	71.64±0.69
Scalar summaries + GBT	7.99±0.95	74.78±1.32	72.70±0.50
<i>CAGE-CAL incremental variants</i>			
Observed graph encoder only	6.97±0.28	81.25±0.88	75.61±0.13
+ iid counterfactual tower	6.78±0.33	81.89±1.47	75.79±0.24
+ Group-level hyperedge stream	6.75±0.37	82.56±1.27	76.08±0.38
+ Calibration-aware objective (full)	5.56 ±0.03	83.61 ±1.34	76.47 ±0.37
Δ vs. base	−1.41	+2.36	+0.86

Table 3: **Component ablation of CAGE-CAL**, averaged over 25 in-distribution cells (percent, mean ± std over 3 rollouts). Δ rows show absolute changes from the previous variant in percentage points. LR and GBT denote logistic regression and gradient-boosted trees.

confidence summaries. Adding the IID counterfactual tower further improves AUROC by 0.64 points, while the group-level hyperedge stream adds another 0.67 points by capturing shared family, role, answer-cluster, and exposure effects. The calibration-aware objective gives the largest ECE reduction, from 6.75 to 5.56, without sacrificing ranking quality. Overall, the gains come from modeling a counterfactual dependence shift, not from simply adding a stronger prediction head.

7.2 Generalization to Held-Out Topologies

We test topology generalization with leave-one-topology-out evaluation. Each fold removes one topology from both training and validation, and evaluates the calibrator on that unseen topology. As shown in Table 2, CAGE-CAL remains stable under this shift. Its mean AUROC is 82.21, close to the in-distribution result of 83.61, and its mean ECE rises only from 5.56 to 6.56. By comparison, DiscoUQ-LLM reaches 72.27 mean AUROC and 15.82 mean ECE. This indicates that scalar disagreement features do not transfer as well when the communication structure changes. The result supports the relational design of CAGE-CAL. Rather than relying on a topology label, it uses communication edges, local failure correlations, answer clusters, and group-level dependency units. These features are defined for any topology, allowing the calibration rule to transfer to unseen communication structures.

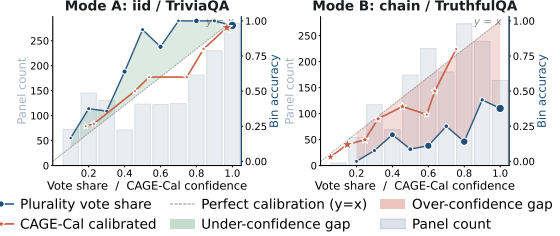


Figure 6: **Failure-mode correction.** Bars show panel counts, and curves show empirical bin accuracy when examples are binned by plurality share or by CAGE-CAL confidence. CAGE-CAL reduces under-confidence in Mode A and over-confidence in Mode B.

7.3 Correcting the Two Failure Modes

Figure 6 tests whether CAGE-CAL corrects the two calibration failures identified earlier. In Mode A, iid/TriviaQA, plurality share is under-confident because wrong answers are dispersed across weak clusters. In Mode B, chain/TruthfulQA, plurality share is over-confident because communication can concentrate agents on the same wrong answer. In both cases, CAGE-CAL moves the reliability curve closer to the perfect calibration line. Thus, the same vote share can receive different confidence depending on how the agreement or disagreement was formed.

8 Conclusion

We investigate confidence calibration in multi-agent LLM systems and show that agreement alone is an unreliable confidence signal. Our analysis across different benchmarks and communication topologies identifies two recurring failure modes: DUC and COC. We propose CAGE-CAL, a counterfactual agent-graph calibration framework that contrasts post-communication graphs with IID counterfactual graphs to separate independent evidence from correlated failure, which improves reliability discrimination with competitive calibration error. We further introduce CAGE-SELECT, which uses calibrated confidence to dynamically select the most reliable topology and improve final panel accuracy. Overall, our results highlight the importance of topology-aware calibration for reliable multi-agent LLM systems.

9 Limitations

A limitation is that agent participation is not optimized. We construct panels from a predefined agent pool, while real systems may benefit from query-adaptive agent selection, where the system decides which agents are most useful for a given query. Another limitation is that communication is not jointly optimized with calibration. We evaluate a fixed set of candidate communication patterns, but practical systems may need adaptive communication routing, where agents decide which interactions are most useful based on intermediate answers and uncertainty. Extending CAGE-CAL to jointly support adaptive agent selection and communication routing is an important future direction (Li et al., 2026a; Zhang et al., 2025c; Shi et al., 2026; Huang et al., 2026).

References

- David Bani-Harouni, Chantal Pellegrini, Paul Stangel, Ege Özsoy, Kamilia Zaripova, Nassir Navab, and Matthias Keicher. 2026. Rewarding doubt: A reinforcement learning approach to calibrated confidence expression of large language models. In *ICLR*.
- Tiejun Chen, Huaiyuan Yao, Jia Chen, Evangelos E. Papalexakis, and Hua Wei. 2026. Every response counts: Quantifying uncertainty of llm-based multi-agent systems through tensor decomposition. In *ACL*.
- Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. 2021. Training verifiers to solve math word problems. In *arXiv*.
- Yilun Du, Shuang Li, Antonio Torralba, Joshua B. Tenenbaum, and Igor Mordatch. 2024. Improving factuality and reasoning in language models through multiagent debate. In *ICML*.
- Yu Feng, Phu Mon Htut, Zheng Qi, Wei Xiao, Manuel Mager, Nikolaos Pappas, Kishalay Halder, Yang Li, Yassine Benajiba, and Dan Roth. 2025. Rethinking llm uncertainty: A multi-agent approach to estimating black-box model uncertainty. In *EMNLP*.
- Yonatan Geifman and Ran El-Yaniv. 2017. Selective classification for deep neural networks. In *NeurIPS*.
- Google. 2025. gemma-3-12b-it. <https://huggingface.co/google/gemma-3-12b-it>.
- Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q Weinberger. 2017. On calibration of modern neural networks. In *ICML*.
- Sirui Hong, Mingchen Zhuge, Jiaqi Chen, Xiawu Zheng, Yuheng Cheng, Ceyao Zhang, Jinlin Wang, Zili Wang, Steven Ka Shing Yau, Zijuan Lin, Liyang Zhou, Chenyu Ran, Lingfeng Xiao, Chenglin Wu, and Jürgen Schmidhuber. 2024. Metagtpt: Meta programming for a multi-agent collaborative framework. In *ICLR*.
- Jiatan Huang, Zheyuan Zhang, Kaiwen Shi, Yanfang Ye, and Chuxu Zhang. 2026. Evolverouter: Co-evolving routing and prompt for multi-agent question answering. *arXiv*.
- Bo Jiang. 2026. Discouq: Structured disagreement analysis for uncertainty quantification in llm agent ensembles. In *arXiv*.
- Mandar Joshi, Eunsol Choi, Daniel S. Weld, and Luke Zettlemoyer. 2017. Triviaqa: A large scale distantly supervised challenge dataset for reading comprehension. In *ACL*.
- Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, Ethan Perez, Nicholas Schiefer, Zac Hatfield-Dodds, Nova DasSarma, Eli Tran-Johnson, Scott Johnston, Sheer El-Showk, Andy Jones, Nelson Elhage, Tristan Hume, Anna Chen, Yuntao Bai, Sam Bowman, Stanislav Fort, and 17 others. 2022. Language models (mostly) know what they know. In *arXiv*.
- Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. 2017. Lightgbm: A highly efficient gradient boosting decision tree. In *NeurIPS*.
- Thomas N. Kipf and Max Welling. 2017. Semi-supervised classification with graph convolutional networks. In *ICLR*.
- Lorenz Kuhn, Yarin Gal, and Sebastian Farquhar. 2023. Semantic uncertainty: Linguistic invariances for uncertainty estimation in natural language generation. In *ICLR*.
- Ananya Kumar, Percy S Liang, and Tengyu Ma. 2019. Verified uncertainty calibration. In *NeurIPS*.
- Ludmila I. Kuncheva. 2004. *Combining Pattern Classifiers: Methods and Algorithms*. Wiley.
- Yukun Li, Sijia Wang, Lifu Huang, and Li-Ping Liu. 2025. Graph-based confidence calibration for large language models. *TMLR*.
- Ziming Li, Jiatan Huang, Xiaoguang Guo, Guilin Wang, and Chuxu Zhang. 2026a. Same signal, opposite meaning: Direction-informed adaptive learning for llm agents. *arXiv*.
- Ziming Li, Xiaoming Wu, Zehong Wang, Jiazheng Li, Yijun Tian, Jinhe Bi, Yunpu Ma, Yanfang Ye, and Chuxu Zhang. 2026b. Graph is a substrate across data modalities. In *ICML*.

- Stephanie Lin, Jacob Hilton, and Owain Evans. 2022a. Teaching models to express their uncertainty in words. *TMLR*.
- Stephanie Lin, Jacob Hilton, and Owain Evans. 2022b. Truthfulqa: Measuring how models mimic human falsehoods. In *ACL*.
- MetaAI. 2024. Llama-3-8b-instruct. <https://huggingface.co/meta-llama/Llama-3.1-8B-Instruct>.
- Microsoft. 2024. phi-4. <https://huggingface.co/microsoft/phi-4>.
- John Platt and 1 others. 1999. Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods. *Advances in large margin classifiers*.
- Dan Qiao, Binbin Chen, Fengyu Cai, Jianlong Chen, Wenhao Li, Fuxin Jiang, Zuzhi Chen, Hongyuan Zha, Tieying Zhang, and Baoxiang Wang. 2026. Episodic gain, aleatoric cost: Uncertainty decomposition in multi-agent debate for math reasoning. In *arXiv*.
- Qwen Alibaba. 2025. Qwen3-8b. <https://huggingface.co/Qwen/Qwen3-8B>.
- Nils Reimers and Iryna Gurevych. 2019. Sentence-bert: Sentence embeddings using siamese bert-networks. In *EMNLP*.
- Kaiwen Shi, Zheyuan Zhang, Zhengqing Yuan, Keerthiram Murugesan, Vincent Galassi, Chuxu Zhang, and Yanfang Ye. 2026. Ng-router: Graph-supervised multi-agent collaboration for nutrition question answering. In *EACL*.
- Mirac Suzgun, Nathan Scales, Nathanael Schärli, Sebastian Gehrmann, Yi Tay, Hyung Won Chung, Aakanksha Chowdhery, Quoc V. Le, Ed H. Chi, Denny Zhou, and Jason Wei. 2023. Challenging big-bench tasks and whether chain-of-thought can solve them. In *ACL*.
- Lei Wang, Wanyu Xu, Yihuai Lan, Zhiqiang Hu, Yunshi Lan, Roy Ka-Wei Lee, and Ee-Peng Lim. 2023a. Plan-and-solve prompting: Improving zero-shot chain-of-thought reasoning by large language models. In *ACL*.
- Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. 2023b. Self-consistency improves chain of thought reasoning in language models. In *ICLR*.
- Yubo Wang, Xueguang Ma, Ge Zhang, Yuansheng Ni, Abhranil Chandra, Shiguang Guo, Weiming Ren, Aaran Arulraj, Xuan He, Ziyang Jiang, Tianle Li, Max Ku, Kai Wang, Alex Zhuang, Rongqi Fan, Xiang Yue, and Wenhui Chen. 2024. Mmlu-pro: A more robust and challenging multi-task language understanding benchmark. In *NeurIPS*.
- Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc Le, and Denny Zhou. 2022. Chain-of-thought prompting elicits its reasoning in large language models. In *NeurIPS*.
- Qingyun Wu, Gagan Bansal, Jieyu Zhang, Yiran Wu, Beibin Li, Erkang Zhu, Li Jiang, Xiaoyun Zhang, Shaokun Zhang, Jiale Liu, Ahmed Hassan Awadallah, Ryen W White, Doug Burger, and Chi Wang. 2024. Autogen: Enabling next-gen llm applications via multi-agent conversation. In *COLM*.
- Ruixin Yang, Dheeraj Rajagopal, Shirley Anugrah Hayati, Bin Hu, and Dongyeop Kang. 2024. Confidence calibration and rationalization for llms via multi-agent deliberation. In *arXiv*.
- Michihiro Yasunaga, Xinyun Chen, Yujia Li, Panupong Pasupat, Jure Leskovec, Percy Liang, Ed H. Chi, and Denny Zhou. 2024. Large language models as analogical reasoners. In *ICLR*.
- Bianca Zadrozny and Charles Elkan. 2002. Transforming classifier scores into accurate multiclass probability estimates. In *KDD*.
- Guibin Zhang, Yanwei Yue, Zhixun Li, Sukwon Yun, Guancheng Wan, Kun Wang, Dawei Cheng, Jeffrey Xu Yu, and Tianlong Chen. 2025a. Cut the crap: An economical communication pipeline for llm-based multi-agent systems. In *ICLR*.
- Guibin Zhang, Yanwei Yue, Xiangguo Sun, Guancheng Wan, Miao Yu, Junfeng Fang, Kun Wang, Tianlong Chen, and Dawei Cheng. 2025b. G-designer: Architecting multi-agent communication topologies via graph neural networks. In *ICML*.
- Jiaxin Zhang, Caiming Xiong, and Chien-Sheng Wu. 2026. Agentic confidence calibration. In *ICML*.
- Zheyuan Zhang, Kaiwen Shi, Zhengqing Yuan, Zehong Wang, Tianyi Ma, Keerthiram Murugesan, Vincent Galassi, Chuxu Zhang, and Yanfang Ye. 2025c. Agentrouter: A knowledge-graph-guided llm router for collaborative multi-agent question answering. *arXiv*.
- Huaxiu Steven Zheng, Swaroop Mishra, Xinyun Chen, Heng-Tze Cheng, Ed H. Chi, Quoc V Le, and Denny Zhou. 2024. Take a step back: Evoking reasoning via abstraction in large language models. In *ICLR*.
- Ziang Zhou, Tianyuan Jin, Jieming Shi, and Qing Li. 2025. Steerconf: Steering llms for confidence elicitation. In *NeurIPS*.
- Xiaochen Zhu, Caiqi Zhang, Yizhou Chi, Tom Stafford, Nigel Collier, and Andreas Vlachos. 2026. Demystifying multi-agent debate: The role of confidence and diversity. In *arXiv*.
- Mingchen Zhuge, Wenyi Wang, Louis Kirsch, Francesco Faccio, Dmitrii Khizbullin, and Jürgen Schmidhuber. 2024. Language agents as optimizable graphs. In *ICML*.

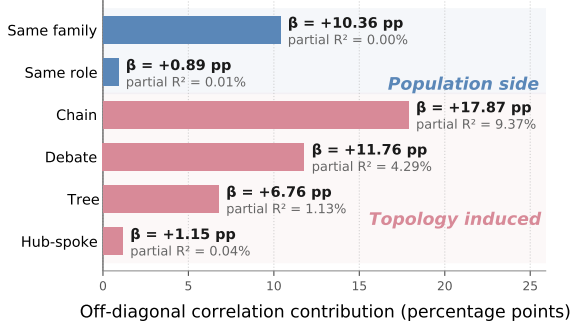


Figure 7: **Sources of agent error correlation.** OLS coefficients on predictors of pair correlation W_{ij} . Communication structure dominates the backbone, which dominates the prompting role.

A Additional Related Work

Graph-based multi-agent design and calibration under dependence. A related line of work represents multi-agent systems as graphs and optimizes their communication structure. GPTSwarm (Zhuge et al., 2024) treats language agents as optimizable graphs, G-Designer (Zhang et al., 2025b) learns communication topologies with graph neural networks, and AgentPrune (Zhang et al., 2025a) removes unnecessary agents or communication links for efficiency. These methods focus mainly on task accuracy, routing, or computational cost. They do not ask how a given topology changes the panel’s joint failure distribution, nor do they produce calibrated confidence for the panel answer. Our work turns graph structure into a calibration signal: given a topology and the resulting panel outputs, we estimate whether the panel’s agreement reflects independent evidence or correlated failure.

B Failure-Mode Analysis and Counterfactual Motivation

B.1 Why Calibration Needs Counterfactual Graph Shifts

In this section, we first explain why confidence calibration requires both the iid graph G_x^0 and the post-communication graph G_x^T . We then provide empirical evidence that the shift between G_x^0 and G_x^T matters for confidence calibration.

B.1.1 Why Calibration Needs the IID and Post-Communication Graphs

Section 3 defines $W(x)$ as the local dependency structure of a panel. We now ask what creates this dependency. If agents provided independent evidence, the off-diagonal entries of $W(x)$ would

Predictor	β (pp)	Partial R^2 (%)
Intercept (baseline iid pair)	35.40	—
<i>Population-side sources</i>		
Same backbone family	+10.36	0.00
Same prompting role	+0.89	0.01
<i>Topology-induced sources (vs. iid baseline)</i>		
is_chain	+17.87	9.37
is_debate	+11.76	4.29
is_tree	+6.76	1.13
is_hub_spoke	+1.15	0.04
Full-model R^2	29.02%	
n pairs	4,325	

Table 4: **Decomposition of agent error correlation W_{ij} .** OLS on 4,325 agent pairs from the 25 cells. β is the coefficient on the 0/1 predictor in percentage points; partial R^2 is the variance lost when dropping that predictor.

be small and weakly structured. Instead, we find two systematic sources of correlation: population-side similarity among agents and communication-induced coupling from the topology. This distinction is central to calibration. A topology-agnostic score can observe the final vote distribution, but it cannot tell whether the distribution was produced by independent evidence or correlated failure.

Where does failure correlation come from?

For each topology and benchmark cell, we aggregate the off-diagonal entries of $W(x)$ and fit an ordinary least squares model over agent pairs:

$$W_{ij}^{(c)} = \beta_0 + \beta_F \mathbf{1}[\text{same family}] + \beta_R \mathbf{1}[\text{same role}] + \sum_{t \in \mathcal{T} \setminus \{\text{iid}\}} \beta_t \mathbf{1}[\text{topology}_c = t] + \epsilon.$$

Figure 7 visualizes the main coefficients, while the full regression results are reported in Table 4. Table 4 decomposes pairwise agent failure correlation into population-side and topology-induced sources. The coefficients do not measure final panel accuracy; instead, they measure how much each factor increases the tendency of two agents to succeed or fail together. Shared backbone families increase pairwise correlation, suggesting common model-side blind spots, while communication topologies such as chain and debate further amplify correlated failures. These results support our motivation for modeling both population-side dependence and topology-induced coupling in multi-agent confidence calibration.

The decomposition gives two observations.

- **First, heterogeneous model families matter more than heterogeneous prompts.** Agents with the same backbone tend to fail together even when they use different prompting roles. In contrast, shared prompting role contributes little once backbone family is controlled. This suggests that prompt diversity can make a panel appear diverse without fully decorrelating its errors.
- **Second, communication topology is a major source of additional dependence.** Chain has the largest topology coefficient, followed by debate and tree. This order follows the amount of peer exposure in each topology: later agents in a chain see many previous outputs, debate agents see many peers in one round, and tree agents receive information through local branches. Hub-spoke is close to the iid baseline because spokes do not see one another. Thus, part of the dependence is inherited from the heterogeneous agent population, while another part is induced by communication.

Motivating the Counterfactual Graph Pair

This split directly motivates the graph pair used by CAGE-CAL. A counterfactual iid graph G_x^0 captures the dependence that would exist without communication. It represents population-side structure, including shared model families, shared biases, and shared prompting conventions. A post-communication graph G_x^T captures the same matched agents after topology T has reshaped their outputs. It contains both population-side dependence and communication-induced dependence. Because G_x^0 and G_x^T are built over the same agent identities, their contrast gives a query-specific view of how communication changes the panel. This is the information that scalar disagreement scores discard. Section 5 turns this counterfactual graph pair into the input representation for CAGE-CAL.

B.2 Communication-Induced Correctness Shifts

Section B.1.1 shows that communication can increase failure correlation among agents. We now examine the same effect at the agent level. Communication can change an agent’s correctness in both directions. It can help an initially wrong agent become correct, but it can also make an initially correct agent become wrong. Mean accuracy only

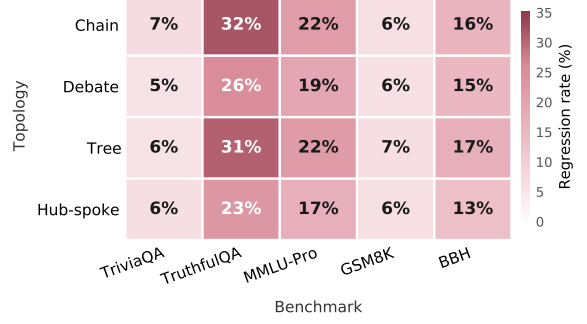


Figure 8: **Per-agent regression rate.** Fraction of iid-correct agents that become wrong under each topology on the same (question, rollout). Substantial on hard benchmarks even in topologies whose mean accuracy is unchanged.

reflects the net balance of these two effects. Calibration depends on how these shifted votes are distributed in the final panel.

Regression and improvement. For agent i , query x , rollout r , and topology τ , we call the agent *communication-regressed* when $c_i^{(\text{iid})}(x, r) = 1$ and $c_i^{(\tau)}(x, r) = 0$. That is, the agent answers correctly when run independently, but answers incorrectly after communication. We call the symmetric case *communication-improved*, when $c_i^{(\text{iid})}(x, r) = 0$ and $c_i^{(\tau)}(x, r) = 1$. For each topology τ and benchmark b , we report both the regression rate $\text{reg}_{\tau, b}$ and the improvement rate $\text{impr}_{\tau, b}$. The iid run provides a matched counterfactual for the same agent on the same query.

Communication can help accuracy while hurting calibration. Figure 8 shows that regression is not rare. On hard benchmarks, a substantial fraction of independently correct agents become wrong after communication. For example, chain on TruthfulQA regresses nearly one third of agents that were correct in the iid setting. Tree shows a similar pattern on TruthfulQA and MMLU-Pro.

At the same time, regression does not necessarily imply lower mean accuracy. Communication can also improve agents that were initially wrong. A topology can therefore be helpful for average accuracy while still injecting many wrong votes into the panel. This distinction is central to calibration. A corrected vote and a regressed vote both appear as confident votes in the final answer distribution. Plurality vote cannot tell whether a larger answer cluster was formed by independent correction or by harmful influence.

Topology	Benchmark	Reg. rate	Impr. rate	ΔAcc (pp)
Debate	TriviaQA	5.44%	39.55%	+7.44
	TruthfulQA	25.60%	11.05%	+3.16
	MMLU-Pro	19.15%	20.24%	+4.16
	GSM8K	5.86%	54.65%	+5.90
	BBH	14.99%	28.68%	+2.78
Chain	TriviaQA	7.05%	35.10%	+5.02
	TruthfulQA	31.73%	9.49%	+0.62
	MMLU-Pro	21.52%	21.78%	+4.10
	GSM8K	6.12%	54.53%	+5.66
	BBH	15.74%	28.92%	+2.43
Hub-spoke	TriviaQA	6.31%	15.86%	+0.04
	TruthfulQA	23.12%	6.04%	-0.24
	MMLU-Pro	17.09%	12.69%	+0.53
	GSM8K	6.03%	24.85%	-0.03
	BBH	13.31%	19.79%	+0.16
Tree	TriviaQA	5.64%	31.45%	+5.02
	TruthfulQA	30.52%	7.51%	-0.26
	MMLU-Pro	21.99%	16.97%	+0.94
	GSM8K	6.62%	43.69%	+2.64
	BBH	17.19%	24.34%	-0.16

Table 5: **Communication-induced regression rates.** *Reg.* = fraction of iid-correct agents that became wrong under the row’s topology on the same (question, roll-out). *Impr.* is the converse. ΔAcc is the net per-agent accuracy change in percentage points.

The key implication is that communication changes the reliability of votes, not only their average correctness. Two topologies can have similar mean accuracy but very different calibration behavior, because their correctness shifts land in different answer clusters. The next section shows how these shifts lead to two opposite calibration failures: diversity-induced under-confidence and communication-induced over-confidence.

Communication Can Both Correct and Corrupt Agents Table 5 analyzes how communication changes individual agent correctness relative to the iid setting. The regression rate measures the fraction of iid-correct agents that become incorrect after communication, while the improvement rate measures the reverse transition from incorrect to correct. Communication can improve many agents, especially on GSM8K, but it can also corrupt initially correct agents, with particularly high regression rates on TruthfulQA. This shows that communication reshapes agent-level failure patterns rather than simply improving accuracy, motivating calibration methods that account for topology-induced dependence.

Topology	Mean ECE
debate	14.59 ± 8.26
tree	14.82 ± 6.09
hub-spoke	17.55 ± 8.02
iid	18.00 ± 8.60
chain	19.48 ± 16.31

Table 6: **Per-topology mean majority-vote ECE** (% , mean $\pm$ std across benchmarks). Chain is worst-calibrated and benchmark-variant (benign on TriviaQA, catastrophic on TruthfulQA).

B.3 Calibration Error Varies Across Communication Topologies

Table 6 reports the majority-vote ECE for each communication topology, averaged across benchmarks. Unlike plurality accuracy, which remains relatively stable across topologies, calibration error varies more substantially. In particular, chain has the highest mean ECE and the largest variance, suggesting that communication structure can make vote-based confidence unreliable even when final-answer accuracy changes little. This motivates our focus on calibrating panel confidence rather than only improving the predicted answer.

B.4 Vote Share Is Not a Reliable Confidence Estimate

Figure 9 shows that plurality vote share has different calibration behavior across benchmarks and communication topologies. Each point compares a vote-share bin with the empirical accuracy of panel predictions in that bin; points on the diagonal indicate perfect calibration. The patterns are highly cell-dependent: on TriviaQA and GSM8K, many points lie above the diagonal, meaning that vote share underestimates correctness, while on TruthfulQA many points lie below the diagonal, meaning that vote share overestimates correctness. Moreover, the same benchmark can behave differently under different topologies, such as iid, debate, chain, hub-spoke, and tree. This shows that vote share alone cannot determine panel confidence; calibration must account for both task domain and the communication structure that produced the vote.

B.5 Agent Error Correlations Vary Across Topologies and Benchmarks

Figure 10 shows that agent correctness correlations vary across model families, communication topologies, and benchmarks. Darker entries indicate that two agents tend to succeed or fail to-

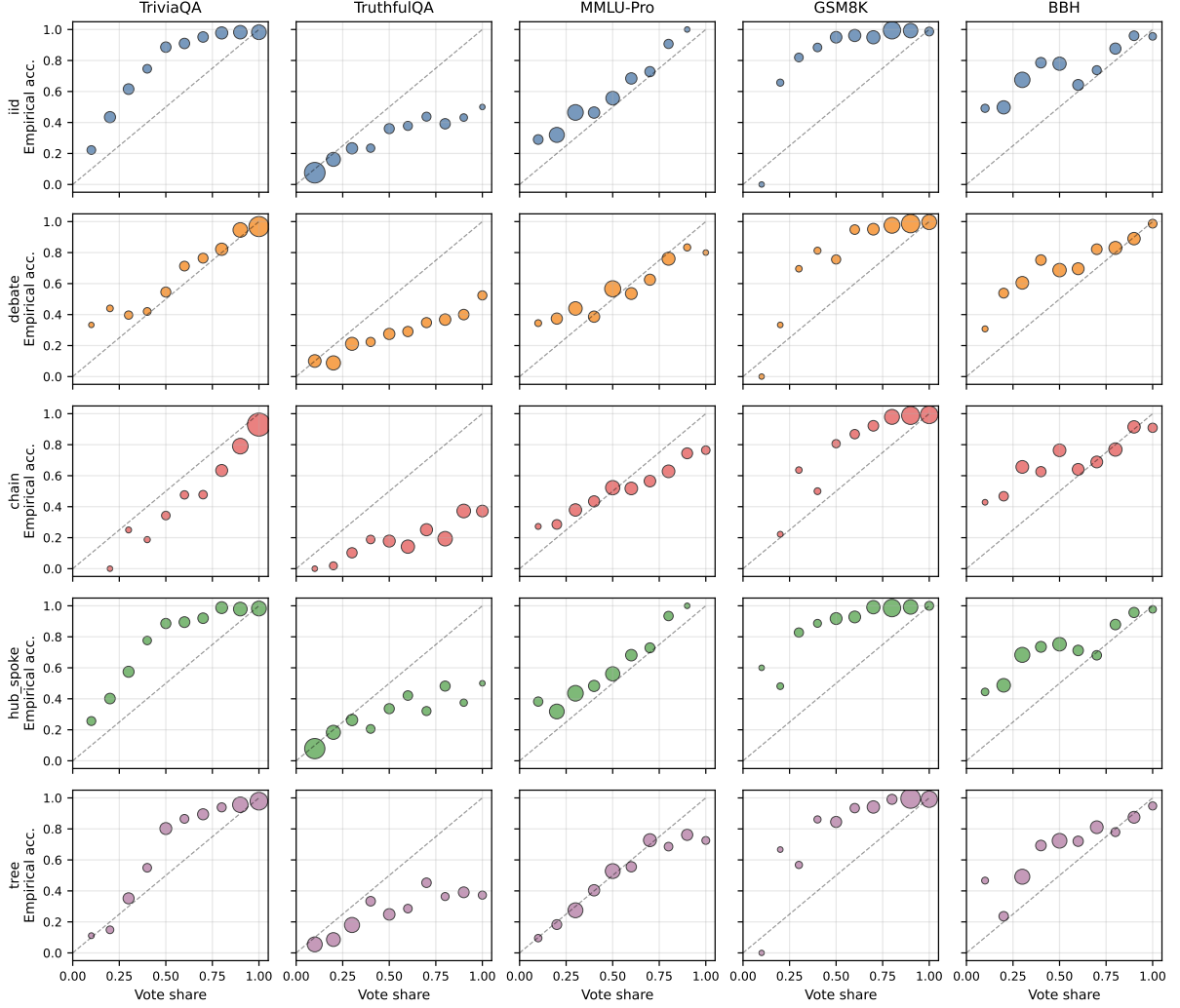


Figure 9: **Reliability diagrams per (topology, benchmark) cell.** Plurality vote share (x) vs. empirical accuracy (y). Bubble area is the bin’s panel count. Points above the identity line are under-confident (Mode A); below, over-confident (Mode B).

gether. Within-family blocks, such as agents sharing the same Gemma, Llama, Phi, or Qwen backbone, are often darker, suggesting shared model-side blind spots. Comparing rows shows that topology also changes the dependence pattern: chain and debate often make the matrices darker than iid, indicating stronger communication-induced coupling. Comparing columns shows that this effect is benchmark-dependent: correlations are much stronger on TriviaQA and TruthfulQA than on MMLU-Pro, GSM8K, or BBH. These patterns show that multi-agent reliability depends jointly on heterogeneous LLM composition, communication structure, and task domain, motivating topology-aware calibration.

C Method Details

C.1 Node Feature Construction

Each node represents one agent. For both G_x^T and G_x^0 , we use the same 23-dimensional node feature schema:

$$v_i(x) = [s_i, p_i, r_i, \mu_i, \sigma_i^2, m_i, n^{-1}, a_i]$$

The confidence score s_i is computed from the agent’s mean answer log probability. We clip the mean log probability to $[-10, 0]$ and linearly map it to a normalized score. The plurality indicator p_i is one when the agent’s normalized answer matches the panel plurality answer. The vote rank r_i is normalized by the number of distinct answer clusters in the panel. The correlation summaries (μ_i, σ_i^2, m_i) are the row mean, row variance, and row maximum of $|W_{ij}(x)|$ over other agents. For

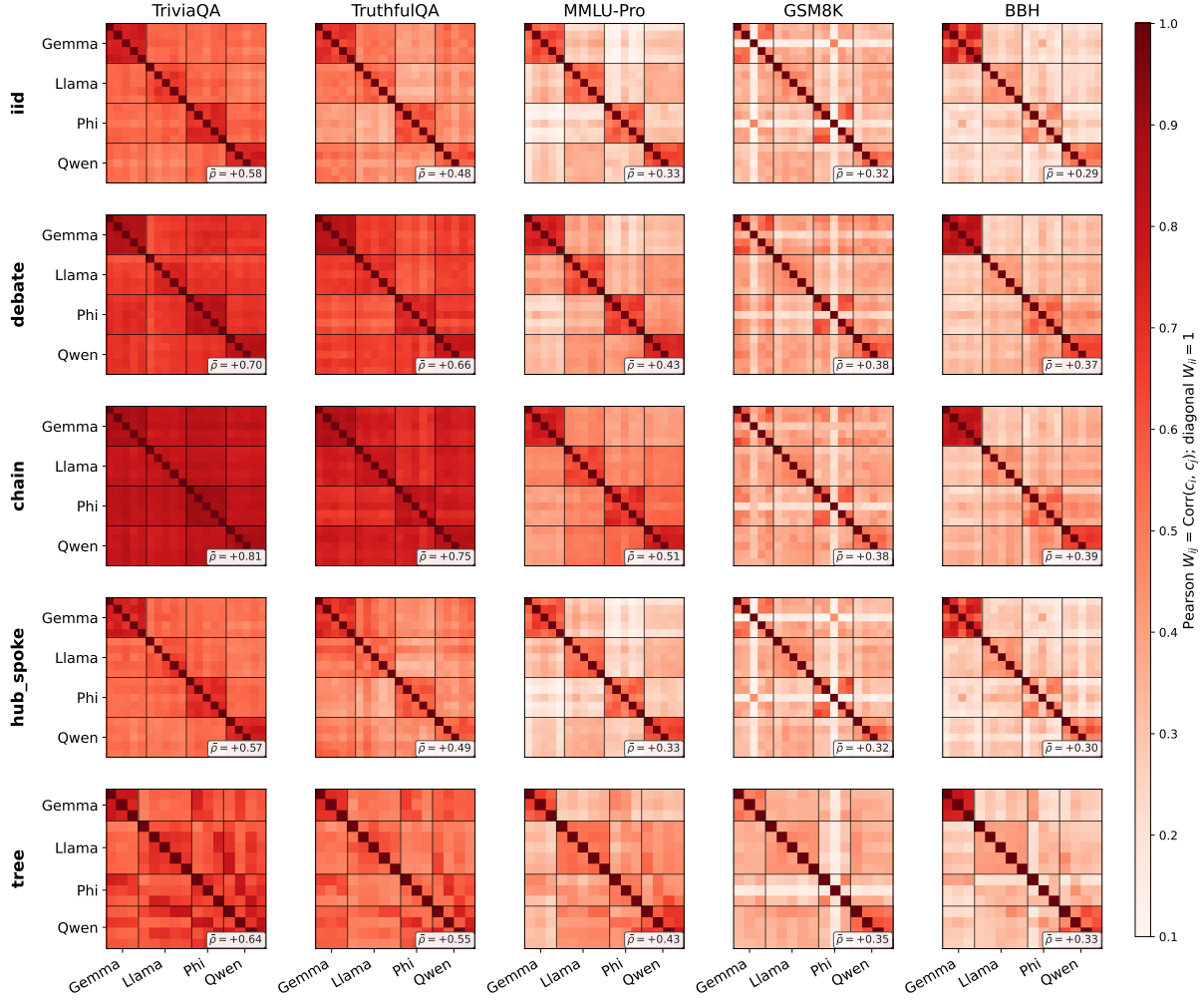


Figure 10: **Per-(topology, benchmark) agent error correlation $\mathbf{W}$** . Each panel is an $N \times N$ Pearson correlation of the binary error indicator c_i across panel agents, with per-query mean residualisation to control for task difficulty. Agents are sorted by (backbone, role) so visual patterns are comparable across topologies.

the observed tower, these summaries are computed from the post communication correlation matrix. For the iid counterfactual tower, they are computed from the iid correlation matrix. Finally, the answer embedding a_i is obtained by encoding the agent’s answer with Sentence-BERT (Reimers and Gurevych, 2019) and reducing the embedding to 16 dimensions by PCA. Thus, each node records both the agent’s individual contribution to the vote and its local dependence pattern with the rest of the panel.

C.2 Instance-Conditional Correlation Estimation

We estimate the correlation from the training split only. For each query, we retrieve the $k = 20$ nearest training queries in Sentence-BERT embedding space and compute the empirical Pearson correlation of agent correctness over this local neigh-

borhood. Pairs with nearly zero variance are assigned correlation zero. Validation and test labels are never used to construct these matrices.

In implementation, we keep an edge if either the agents are directly connected by communication or the absolute local correlation exceeds 0.05. For the iid counterfactual graph, only the correlation criterion is used. This keeps the graph focused on meaningful dependence while preserving direct communication links. If no edge remains after thresholding, we use a fully connected graph without self loops to avoid degenerate isolated graphs.

D Implementation Details

D.1 Benchmarks

We evaluate on five English-language benchmarks chosen to span distinct skill profiles: short-form factual recall (TriviaQA), truthfulness in the

face of common misconceptions (TruthfulQA), broad expert-level knowledge (MMLU-Pro), grade-school numerical reasoning (GSM8K), and a diverse battery of hard reasoning tasks (BBH).

TriviaQA (Joshi et al., 2017). A large-scale closed-book QA dataset of factoid trivia questions sourced from quiz league competitions. Gold answers are short strings, and an official multi-answer alias set is provided to accommodate surface-form variation at grading time.

TruthfulQA (Lin et al., 2022b). 817 questions designed to probe whether models reproduce common human misconceptions or imitative falsehoods. We use the open-ended generation setting, in which each question ships with reference sets of correct and incorrect answers used to grade free responses.

MMLU-Pro (Wang et al., 2024). An enhanced version of MMLU with up to ten answer options per question and stronger distractors than the original, covering 14 subject categories. Questions are presented in standard A–J multiple-choice format and graded by the selected letter.

GSM8K (Cobbe et al., 2021). Grade-school math word problems requiring multi-step arithmetic reasoning, each with a single numerical gold answer. Grading compares numerical equality of the model’s parsed answer against the canonical gold value.

BBH (Suzgun et al., 2023). BIG-Bench Hard, a collection of 23 subtasks drawn from BIG-Bench on which prior LLMs underperformed humans. Subtasks vary in answer format (multiple choice, yes/no, free-form), and grading dispatches to a subtask-specific judge.

D.2 Agent designs

Backbones. We use four open-weight LLM backbones drawn from four distinct model families: QWEN3-8B (Qwen Alibaba, 2025), LLAMA-3.1-8B (MetaAI, 2024), GEMMA-3-12B (Google, 2025), and PHI-4 (Microsoft, 2024). Using four independently pretrained families is intended to expose realistic disagreement: agents sometimes fail in correlated ways because they share training data or recipe, and sometimes disagree because they come from different pipelines.

Prompting roles. Each backbone is paired with one of five atomic single-pass reasoning

roles, each adapted verbatim from its source paper and summarized in Appendix G: DIRECT (zero-shot, no reasoning shown), COT (Wei et al., 2022), PLAN_SOLVE (Wang et al., 2023a), STEP_BACK (Zheng et al., 2024), and ANALOGICAL (Yasunaga et al., 2024). Restricting each agent to a single LLM call and a single role keeps the role axis and the topology axis orthogonal, so any multi-round or multi-agent behavior comes from the topology rather than from a role internally hiding a sub-population.

Canonical population. An *agent* is a (backbone, prompting role) pair. The full 4×5 cross product yields 20 deterministically ordered atomic agents that constitute our canonical population, and the *same* population is reused across every topology so that any cross-topology difference can be attributed to communication structure rather than to a different mix of agents.

D.3 Communication Topology Design

We compare five communication topologies that span the spectrum from no communication to fully connected, and from peer-to-peer to hierarchical aggregation. All topologies share the same agent population, the same per-agent sampling temperature, the same maximum decode length, and the same plurality-vote aggregator at the end. Each topology is run with 3 rollouts per query for reproducibility.

IID (no communication). The N agents answer the query independently and in parallel. This is the standard self-consistency setting and serves as the counterfactual baseline against which the four communicating topologies are compared. The iid rollout is also the source of the counterfactual graph G_x^0 used by CAGE-CAL.

DEBATE (two-round full mesh). Round 1 is identical to iid. In round 2, every agent receives a formatted block listing all other agents’ round-1 answers and is invited to revise. The final panel is the set of round-2 answers, so every agent has been exposed to every peer exactly once before committing.

CHAIN (sequential pipeline). The N agents are arranged in a per-rollout shuffled order $a_1 \rightarrow a_2 \rightarrow \dots \rightarrow a_N$. Agent a_i sees the formatted answers of $a_1, \dots, a_{i-1}$ before producing its own. The final panel is the set of all N answers; the last agent has the largest peer context, the first agent has none.

HUB-SPOKE (centralized aggregation). The $N-1$ spoke agents answer the query independently and in parallel. The hub then sees a formatted block of all spoke answers labelled as “worker” contributions and produces its own answer. The final panel includes the hub and all spokes, with the hub’s answer typically dominating the plurality vote.

TREE (hierarchical aggregation). A complete binary tree of depth L aggregates leaf answers upward. At the bottom layer, 2^L leaf agents answer independently. Each internal node sees the answers of its two children and produces its own answer; the root produces the panel’s most informed answer. All nodes’ answers are retained for the panel. With our canonical $N=15$ subset, the tree has depth $L=3$ and shape $8+4+2+1$.

D.4 Baselines

We compare CAGE-CAL against four families of baselines. The first family applies classical post-hoc calibration to the raw plurality vote share. The second asks an LLM judge to score the panel directly. The third trains a learned calibration head on validation data. The fourth reports ranking-only uncertainty scores that do not target calibrated probabilities.

D.4.1 Post-hoc plurality calibrators

Methods in this family use the plurality vote share as their raw confidence signal (Kuncheva, 2004). This is the fraction of agents that support the panel’s most-voted answer. The four methods differ only in how this scalar is reshaped into a final probability. *Plurality share* uses the raw value with no learnable parameters and serves as the simplest calibration reference. + *Platt scaling* (Platt et al., 1999) fits a parametric logistic mapping on the validation split. + *Isotonic regression* (Zadrozny and Elkan, 2002) instead fits a non-parametric monotone mapping. It can correct non-sigmoidal miscalibration but introduces more variance. + *Scaling-binning* (Kumar et al., 2019) chains a parametric scaler with empirical bin-mean replacement and provides finite-sample ECE guarantees.

D.4.2 LLM-elicited confidence estimators

These baselines test whether panel correctness can be inferred from the final agent responses alone. We prompt an LLM judge to return a probability that the plurality answer is correct. The judge sees the question, the agent answers, and the plurality answer, but no graph or per-token statistics.

Full prompts appear in Appendix G. *LLM-Cal (no topology)* queries the judge directly. *LLM-Cal (+topology)* additionally injects a one-line description of the panel’s communication topology into the prompt. This isolates whether the judge can exploit topology information when given it explicitly. *Collaborative Calibration* (Yang et al., 2024) prompts the judge to silently simulate a multi-expert deliberation before emitting a single consensus probability.

D.4.3 Trained calibrators

These baselines train a calibration head on validation panels and target plurality-answer correctness. To make the comparison fair, all three trained baselines receive the same per-benchmark post-hoc protocol as our method. *Scalar + GBT* (Ke et al., 2017) is a strong feature-based baseline. It trains a gradient-boosted decision tree on hand-crafted scalar panel statistics, with no relational encoding. *GraphCal* (Li et al., 2025) adapts a graph-based calibrator to the panel setting. It encodes the observed agent graph with a GCN (Kipf and Welling, 2017), but it omits the counterfactual iid view, the hyper-edge stream, and the failure-correlation edges used by CAGE-CAL. *DiscoUQ-LLM* (Jiang, 2026) is a disagreement-feature baseline. It trains a head on a small set of panel-level disagreement summaries.

D.4.4 Ranking-only UQ scores

This family does not produce calibrated probabilities. It only yields scalar uncertainty estimates that can rank panels by reliability, so we report only AUROC and AUARC for these methods. *Answer entropy* (Kuhn et al., 2023) is the Shannon entropy of the panel’s distribution over distinct agent answers. *Average log probability* (Kadavath et al., 2022) averages each agent’s mean per-token log-probability across the panel. It uses only signals that the decoder already provides. *DiverseAgentEntropy* (Feng et al., 2025) extends answer entropy by first diversifying the agents that produce the answer distribution. *MATU* (Chen et al., 2026) arranges the panel’s per-agent answer distribution into a low-rank tensor and uses the PARAFAC2 reconstruction residual as the uncertainty score.

D.5 Training Details

Data splits. For each of the five benchmarks we sample 500 problems with a fixed seed and deterministically split them 60:20:20 into train, validation, and test partitions. The same split is reused

Method	TriviaQA		TruthfulQA		MMLU-Pro		GSM8K		BBH		Mean	
	AUROC $\uparrow$	AUARC $\uparrow$	AUROC $\uparrow$	AUARC $\uparrow$	AUROC $\uparrow$	AUARC $\uparrow$	AUROC $\uparrow$	AUARC $\uparrow$	AUROC $\uparrow$	AUARC $\uparrow$	AUROC $\uparrow$	AUARC $\uparrow$
Plurality vote	81.89 $\pm$ 0.79	93.09 $\pm$ 1.14	71.16 $\pm$ 3.89	31.30 $\pm$ 1.22	61.78 $\pm$ 0.80	53.02 $\pm$ 0.40	84.01 $\pm$ 2.22	97.87 $\pm$ 0.22	66.15 $\pm$ 1.49	84.22 $\pm$ 0.46	72.99 $\pm$ 1.52	71.90 $\pm$ 0.07
Answer entropy	80.19 $\pm$ 0.67	92.82 $\pm$ 1.11	70.83 $\pm$ 2.97	31.36 $\pm$ 0.93	57.92 $\pm$ 0.98	51.01 $\pm$ 0.70	82.60 $\pm$ 0.99	97.72 $\pm$ 0.08	67.98 $\pm$ 1.30	84.94 $\pm$ 0.52	71.90 $\pm$ 1.11	71.57 $\pm$ 0.06
Avg-logprob	69.83 $\pm$ 2.81	86.96 $\pm$ 0.74	60.31 $\pm$ 0.41	27.65 $\pm$ 1.01	43.65 $\pm$ 2.08	27.33 $\pm$ 0.40	57.87 $\pm$ 2.32	75.41 $\pm$ 0.48	46.11 $\pm$ 1.90	49.77 $\pm$ 1.47	55.56 $\pm$ 0.91	53.42 $\pm$ 0.82
DAE	80.19 $\pm$ 0.66	92.82 $\pm$ 1.11	70.85 $\pm$ 3.01	31.34 $\pm$ 0.95	57.91 $\pm$ 0.94	51.01 $\pm$ 0.72	82.61 $\pm$ 1.00	97.72 $\pm$ 0.08	67.96 $\pm$ 1.31	84.94 $\pm$ 0.53	71.91 $\pm$ 1.12	71.57 $\pm$ 0.05
MATU	71.23 $\pm$ 2.55	89.73 $\pm$ 0.33	59.80 $\pm$ 2.57	26.88 $\pm$ 1.82	61.39 $\pm$ 0.96	52.20 $\pm$ 0.24	56.68 $\pm$ 1.32	95.45 $\pm$ 0.19	48.81 $\pm$ 2.91	73.89 $\pm$ 0.19	59.58 $\pm$ 1.68	67.63 $\pm$ 0.42
CAGE-CAL (ours)	86.12 $\pm$ 0.78	93.02 $\pm$ 0.26	79.66 $\pm$ 2.43	38.76 $\pm$ 1.44	77.74 $\pm$ 0.75	61.54 $\pm$ 0.88	84.04 $\pm$ 3.55	97.51 $\pm$ 0.51	90.48 $\pm$ 0.81	91.52 $\pm$ 0.74	83.61 $\pm$ 1.34	76.47 $\pm$ 0.37

Table 7: **Heuristic UQ baselines and CAGE-CAL: per-benchmark AUROC and AUARC.** Mean $\pm$ std over 3 rollouts, averaged across the 5 topologies. ECE not applicable to the five ranking-only methods; CAGE-CAL’s ECE is in Table 1. CAGE-CAL AUROC matches Table 1.

across every topology, every method, and every seed, so a question is never seen in training under one topology and held out under another. Because each problem is rolled out through every topology with three independent rollouts, the resulting panel pool contains roughly 37,500 panels in total. The training split fits the encoder and the calibration head, the validation split fits the post-hoc calibrators and selects model checkpoints, and the test split is used only for final reporting. For the leave-one-topology-out experiments the same per-benchmark splits are kept fixed and only the topology used during training is varied.

Optimization. We train the encoder for 15 epochs with AdamW, an initial learning rate of 2×10^{-3} , weight decay of 3×10^{-4} , and a cosine learning-rate schedule. Each batch contains 128 panels and gradients are clipped to a global norm of 1.0. The training loss combines binary cross-entropy on plurality-answer correctness with a Brier auxiliary term weighted by 0.4, and the target is label-smoothed with $\alpha = 0.05$ to prevent the model from collapsing to extreme probabilities on high-agreement panels. Model selection on the validation split uses the composite score $\text{AUROC} - 0.5 \cdot \text{ECE}$, with an early-stopping patience of 5 epochs. We train an ensemble of 10 random seeds and average the predicted probabilities, then apply a per-benchmark Beta calibration and scaling-binning post-hoc and blend the two outputs with equal weight.

Computational cost. The bulk of the wall-clock cost in our pipeline lies in generating the multi-agent panels with vLLM-served open-weight LLMs, which we cache and reuse across all methods. Training CAGE-CAL on top of the cached panels is comparatively cheap: a single 15-epoch training run completes in roughly 5–10 minutes on a single NVIDIA A100, and the full 10-seed ensemble plus per-benchmark post-hoc fits in well

under two hours on the same hardware.

E Additional Calibration and Uncertainty Results

E.1 Ranking-Only UQ Comparison

Table 7 compares CAGE-CAL with heuristic uncertainty baselines across benchmarks. Plurality vote and answer entropy rely only on the final answer distribution, while log-probability and other uncertainty scores do not explicitly model agent dependencies. CAGE-CAL achieves the best mean AUROC and AUARC, showing that topology-aware dependency features provide a stronger signal for estimating whether the panel answer is correct.

E.2 Brier Score Comparison

Table 8 compares probability calibration using Brier score, where lower values indicate better calibrated confidence. The baselines cover three categories: post-hoc plurality calibrators, LLM-elicited confidence estimators, and trained calibrators. CAGE-CAL achieves the lowest Brier score on every benchmark and the best mean score overall, reducing the average Brier score from the strongest baseline, DiscoUQ-LLM, from 14.37 to 11.19. This suggests that explicitly modeling agent dependencies and communication topology provides a stronger calibration signal than calibrating vote share, eliciting verbalized confidence, or using hand-crafted disagreement features alone.

F Selection, Generalization, and Robustness

F.1 CAGE-SELECT: Confidence-Routed Topology Selection

Table 9 evaluates whether the calibrated confidence produced by CAGE-CAL can be used to select, for each query, the communication topology whose panel answer is most likely to be correct. We refer to this routing procedure as CAGE-SELECT.

Method	TriviaQA	TruthfulQA	MMLU-Pro	GSM8K	BBH	Mean
<i>Post-hoc plurality calibrators</i>						
Plurality share	12.12 $\pm$ 0.23	24.52 $\pm$ 0.90	25.01 $\pm$ 0.18	9.61 $\pm$ 0.21	23.89 $\pm$ 0.11	19.03 $\pm$ 0.19
+ Platt	10.91 $\pm$ 0.06	16.36 $\pm$ 0.76	24.30 $\pm$ 0.19	4.37 $\pm$ 0.18	17.97 $\pm$ 0.02	14.78 $\pm$ 0.21
+ Isotonic	10.89 $\pm$ 0.18	16.45 $\pm$ 0.93	24.76 $\pm$ 0.42	4.50 $\pm$ 0.27	17.73 $\pm$ 0.29	14.87 $\pm$ 0.35
+ Scaling-bin.	12.29 $\pm$ 0.09	16.06 $\pm$ 0.30	24.21 $\pm$ 0.02	4.54 $\pm$ 0.01	17.98 $\pm$ 0.11	15.02 $\pm$ 0.03
<i>LLM-elicited confidence estimators</i>						
LLM-Cal (no topo)	10.11 $\pm$ 0.09	46.90 $\pm$ 0.34	35.46 $\pm$ 1.36	10.19 $\pm$ 0.04	19.37 $\pm$ 0.69	24.41 $\pm$ 0.45
LLM-Cal (+topo)	10.75 $\pm$ 0.07	46.51 $\pm$ 0.24	34.60 $\pm$ 0.52	8.96 $\pm$ 0.56	19.40 $\pm$ 0.09	24.04 $\pm$ 0.04
Collab. Cal.	10.32 $\pm$ 0.32	44.21 $\pm$ 0.63	33.71 $\pm$ 0.97	8.41 $\pm$ 0.74	18.26 $\pm$ 0.33	22.98 $\pm$ 0.17
<i>Trained calibrators</i>						
Scalar + GBT	11.43 $\pm$ 0.62	17.49 $\pm$ 1.05	25.96 $\pm$ 0.75	4.87 $\pm$ 0.05	18.11 $\pm$ 0.05	15.57 $\pm$ 0.22
GraphCal	12.29 $\pm$ 0.35	28.32 $\pm$ 0.77	26.28 $\pm$ 0.08	9.25 $\pm$ 0.03	20.18 $\pm$ 0.02	19.26 $\pm$ 0.10
DiscoUQ-LLM	10.47 $\pm$ 0.19	15.31 $\pm$ 0.61	24.02 $\pm$ 0.03	4.46 $\pm$ 0.09	17.57 $\pm$ 0.14	14.37 $\pm$ 0.16
CAGE-CAL (ours)	9.55 $\pm$ 0.80	13.57 $\pm$ 0.53	19.26 $\pm$ 0.46	4.33 $\pm$ 0.22	9.26 $\pm$ 0.48	11.19 $\pm$ 0.50

Table 8: **Brier score ($\downarrow$) per benchmark.** Mean $\pm$ std over 3 rollouts, averaged across the 5 topologies. Reported only for methods that produce a probability natively. CAGE-CAL attains the lowest Brier on every benchmark and the lowest Mean.

Strategy	TriviaQA	TruthfulQA	MMLU-Pro	GSM8K	BBH	Mean
<i>Fixed topology</i>						
iid	83.67	20.00	41.50	94.67	72.51	62.41
debate	84.01	23.67	46.94	94.67	75.95	64.98
chain	82.65	20.67	46.26	94.00	75.95	63.83
hub-spoke	84.01	20.33	42.86	95.33	72.85	63.02
tree	82.99	19.67	41.50	95.00	70.10	61.80
Per-bench best fixed	84.01	23.67	46.94	95.33	75.95	65.18
<i>Selection w/o learned confidence</i>						
Majority over topologies	84.69	19.00	40.82	95.33	75.26	62.95
Highest plurality share	84.35	22.33	44.56	95.33	78.69	64.98
Highest mean log-prob	84.69	22.33	40.82	94.67	74.23	63.29
<i>Confidence-routed</i>						
CAGE-SELECT (ours)	84.69	24.00	50.34	95.33	81.79	67.23
Oracle topology	88.10	31.67	65.31	96.33	86.25	73.43

Table 9: **Confidence-routed topology selection.** Mean accuracy on the 1,479 matched- N test groups (one per query, five candidate panels). *Per-bench best fixed* picks each benchmark’s best topology on validation.

Why per-query routing. The main results show that no single communication topology is best on every query. Some questions are answered most reliably under iid, where independent agents avoid the herding that arises when they communicate. Others benefit from debate or hub-spoke, where peer exchange resolves ambiguity that a lone agent would miss. A *fixed*-topology system is therefore always sub-optimal in expectation: it commits, at design time, to a single communication structure regardless of which structure is appropriate for the specific input. CAGE-SELECT replaces that design-time commitment with a run-time choice driven by panel-level confidence.

Routing procedure. For each query x , CAGE-SELECT runs the panel under all candidate topologies, producing a plurality answer $\hat{y}(x, T)$ and a calibrated correctness probability $\hat{p}(x, T)$ from CAGE-CAL for each T . It then returns the an-

swer of $T^*(x) = \arg \max_T \hat{p}(x, T)$. Because $\hat{p}$ is trained as a panel-level correctness probability with the same target across topologies, the values are directly comparable and the routing reduces to a single $\arg \max$ over a small candidate pool. We compare CAGE-SELECT against each fixed topology, an oracle that always picks the topology carrying the correct answer, and heuristic routers that select by plurality share or mean per-token log-probability. CAGE-SELECT consistently beats every fixed topology and every heuristic rule and closes a substantial fraction of the oracle gap. The heuristics fail in characteristic ways: plurality share is fooled by communication-induced herding that inflates agreement without improving accuracy, and mean log-probability rewards confident generation regardless of whether the panel agrees. A calibrated panel-level confidence is needed to navigate both failure modes, and the same head that

Held-out topology	GraphCal		DiscoUQ-LLM		CAGE-CAL (ours)	
	AUROC	AUARC	AUROC	AUARC	AUROC	AUARC
iid	67.67	69.63	74.89	71.45	84.15	75.23
debate	69.95	71.59	70.32	71.67	80.49	76.60
chain	67.98	70.11	66.97	70.11	82.55	77.32
hub-spoke	70.80	71.87	73.51	72.44	80.12	74.71
tree	73.58	70.63	75.66	70.72	83.74	74.92
Mean	70.00	70.77	72.27	71.28	82.21	75.76

Table 10: **LOTO per-held-out breakdown** (percent, AUROC and AUARC). Each row trains on the other four topologies and tests on the held-out fifth. The Mean row reproduces Table 2’s Mean column.

flags unreliable panels can therefore double as an inference-time selector that turns topology choice into a per-query decision.

F.2 Panel-Size Robustness

Table 10 evaluates leave-one-topology-out generalization, where each row trains the calibrator on four communication topologies and tests it on the held-out topology. CAGE-CAL consistently outperforms GraphCal and DiscoUQ-LLM in both AUROC and AUARC across all held-out topologies, showing that its graph-based dependency modeling generalizes beyond the topologies observed during training. This suggests that CAGE-CAL learns reusable structural signals of agent dependence rather than merely fitting topology-specific patterns.

F.3 Main Structural Findings Are Robust to Panel Size

Table 11 evaluates whether our structural findings are robust to the number of agents in the panel by comparing matched $N = 10$ panels with the natural $N = 20$ setting. Increasing the panel size improves standard accuracy anchors, such as iid plurality accuracy and the any-agent-correct oracle, but the key structural quantities remain stable. In particular, the OLS coefficients on pairwise agent dependence, including same backbone family and topology-induced effects such as chain and debate, change by less than about one percentage point. The peak regression rates on TruthfulQA are also nearly unchanged. These results suggest that the observed population-side dependence and topology-induced coupling are not artifacts of a particular panel size.

Quantity	$N=10$	$N=20$	Δ
<i>Accuracy anchors (%)</i>			
Single-agent baseline	54.13	54.72	+0.59
iid plurality	60.25	64.34	+4.09
Best topology per cell	63.00	66.40	+3.41
Any-agent-correct oracle	84.68	88.08	+3.39
Ensembling gain (iid – single)	+6.12	+9.62	+3.50
Communication gain (best – iid)	+2.75	+2.06	-0.69
Aggregation gap (oracle – best)	+21.69	+21.67	-0.01
<i>OLS β on $\mathbf{W}_{ij}$ (pp)</i>			
Same backbone family	+10.17	+10.36	+0.19
Same prompting role	+1.02	+0.89	-0.14
is_chain	+18.38	+17.87	-0.51
is_debate	+12.19	+11.76	-0.43
is_tree	+6.20	+6.76	+0.55
is_hub_spoke	+1.46	+1.15	-0.31
<i>Peak regression rate (%)</i>			
chain / TruthfulQA	32.45	31.73	-0.72
tree / TruthfulQA	31.18	30.52	-0.66

Table 11: **N -scaling robustness.** Headline quantities under matched- $N=10$ vs. natural- $N=20$. Structural findings move by less than ± 1 pp.

G Prompt Design

Agent role prompts Each agent in a panel is parameterized by one of five atomic reasoning roles whose prompts are taken from the originating papers; a single shared format guard is appended so that downstream parsing of the "Answer: $\langle X \rangle$ " terminator is uniform across roles and benchmarks. A role is restricted to one LLM call and may not internally simulate multiple personas, which keeps the role axis orthogonal to the topology axis. Agent prompting roles used to generate panels are shown in Figure 11.

LLM-elicited baseline prompts For completeness we report the prompt used by the LLM-Cal baseline (Section 6.2), the zero-shot LLM-elicited calibrator that asks a frozen LLM to map (question, panel answers, plurality) to a correctness probability. The optional +topo variant additionally injects a one-line description of the panel’s communication topology; the exact label mapping is shown in the same box. The prompt used for LLM-Cal is shown in Figure 12.

H Use of AI Assistants

AI assistants were used only for language polishing.

Agent prompting roles
Shared format guard (<i>appended to every role below</i>): Output the final answer at the end as exactly one line: "Answer: <your short answer>"
Role 1: DIRECT (zero-shot baseline) Answer the question directly. Do not show reasoning.
Role 2: COT (Wei et al., 2022) Let's think step by step. Reason through the problem, then commit to a final answer.
Role 3: PLAN_SOLVE (Wang et al., 2023a) First, understand the problem and devise a brief plan in 2-4 steps. Then carry out the plan to solve the problem.
Role 4: STEP_BACK (Zheng et al., 2024) Take a step back. State the high-level concept, principle, or category that this problem falls under. Then use that principle to solve the specific problem.
Role 5: ANALOGICAL (Yasunaga et al., 2024) Recall 2-3 analogous problems you have seen before. Briefly describe each in one sentence. Then use what you learned from those analogies to solve this problem.

Figure 11: **Agent prompting roles used to generate panels.** Five atomic reasoning roles parameterize each agent, adapted verbatim from the cited source papers. A role must not internally simulate multiple personas or multiple LLM calls; this keeps role and topology orthogonal.

LLM-Cal (zero-shot LLM-elicited calibrator)
System prompt: You are a calibration assistant. Given a question and N candidate answers from N language model agents, plus the plurality (most-voted) answer, estimate the probability that the plurality answer is correct. Respond with a single decimal number between 0 and 1, no other text.
User prompt template: Question: {query} Panel topology: {topology_description} # only in the +topo variant Plurality answer: {plurality} All agent answers: 1. {answer_1} 2. {answer_2} ... N. {answer_N} Probability the plurality answer is correct (0-1):
Topology descriptions (used by the +topo variant): iid → "iid (independent)" debate → "debate (full-mesh cross-critique)" chain → "chain (sequential)" hub_spoke → "hub-spoke (centralized aggregator)" tree → "tree (hierarchical aggregation by depth)"

Figure 12: **LLM-Cal: zero-shot LLM-elicited calibration baseline.** The model is asked to map (question, panel answers, plurality) to a single probability that the plurality answer is correct. The optional +topo variant additionally injects a one-line description of the panel's communication topology into the user prompt. CAGE-Cal itself does not issue any LLM prompts at inference time.